

# The statistics of soaring flight

Displacement scaling across terrain, circuit types and wing classes

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OFFSITE  
2026

EConophysics Lab

CFM

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Louis  
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# The soaring flight cycle

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## Transition

Glide, lose height

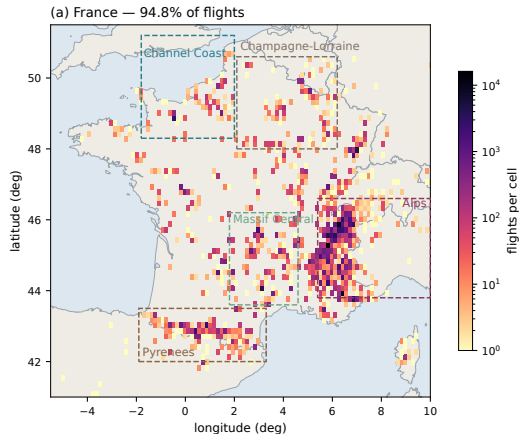
## Climb

Circle, regain height

## Search

Find rising air

Thermals let pilots cover long distances without an engine



## FFVL distance competition

186,052

Paraglider tracklogs  
1999-2026

6,716

Hang-glider tracklogs  
2001-2026

Regional boxes

- Channel Coast
- Champagne-Lorraine
- Massif Central
- Alps
- Pyrenees

IGC trajectories + metadata: time, position, altitude, site, circuit and wing class

## Mean-square displacement (MSD)

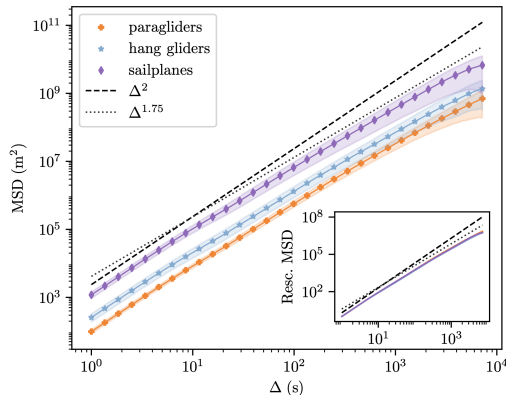
- 1 Along each flight  $f$ , of duration  $T_f$

$$m_f(\tau) = \frac{1}{T_f - \tau} \int_0^{T_f - \tau} |\mathbf{x}_f(t + \tau) - \mathbf{x}_f(t)|^2 dt$$

- 2 Over the  $N$  flights:  $M_2(\tau) = \frac{1}{N} \sum_f m_f(\tau)$

$$M_2(\tau) \propto \tau^{2H}: \quad H = \frac{1}{2} \text{ diffusive, } H = 1 \text{ ballistic}$$

Vilpellet et al.: French flights, 2016–2021



**$M_2(\tau) \propto \tau^{2H}$  with  $H \approx 0.88$ : superdiffusive over three decades,  $10^1$ – $10^4$  s**

# From raw tracklogs to the fixed cohort $C_{10000}$

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## Preprocessing

Clean the recorded flight



Selection and coordinates



Uniform time grid



Savitzky–Golay smoothing



Retained (of downloads)

**156,406**

paragliders, 84%

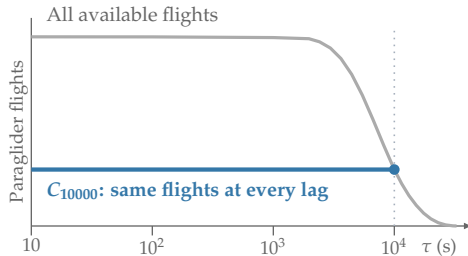
**6,094**

hang gliders, 91%



## Fixed cohort $C_{10000}$

Segments of at least 12,500 s ( $\approx 3.5$  h)



In  $C_{10000}$  (of retained)

**46,273**

paragliders, 30%

**2,326**

hang gliders, 38%

# Which flight characteristics shape MSD and $H$ ?

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## Circuit

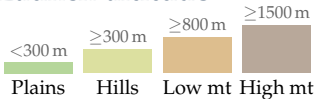


open



closed

## Launch altitude



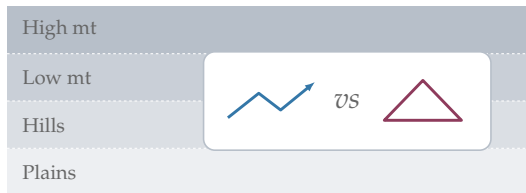
## Wing class

**Beginners** EN A/B/C  
**Experts** EN D/CCC

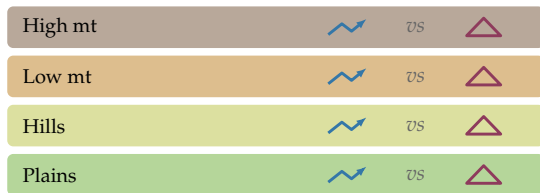
## Weather

*not observed*

**Marginal** across all altitude classes



**Stratified** within each altitude class

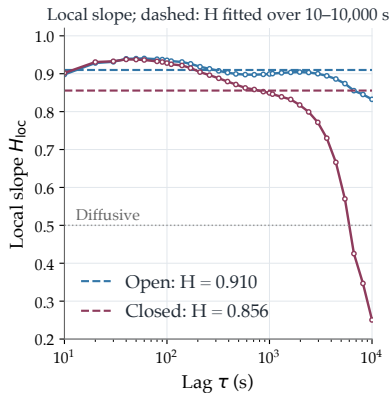
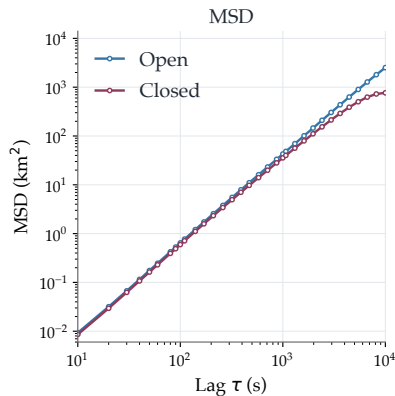


**First compare groups overall, then within the same altitude class.**

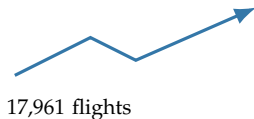
Altitude class set by  $z_0$ , the GNSS altitude of the first fix after trimming.

# Open vs closed: how do MSD and $H$ differ?

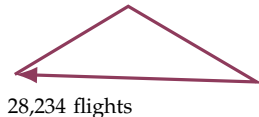
7



Open circuits



Closed circuits

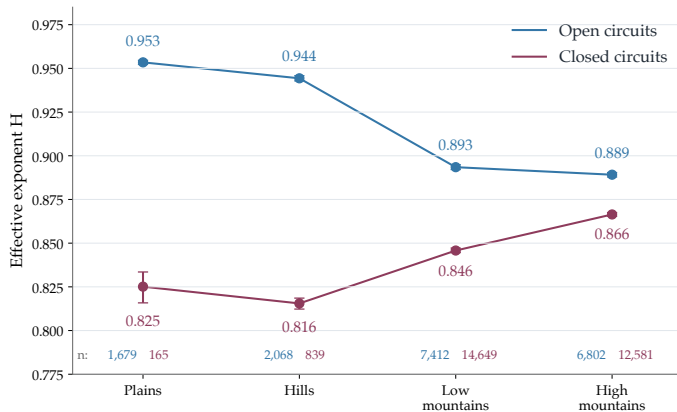


$$H_{\text{open}} - H_{\text{closed}} = 0.054 [0.053, 0.056] > 0 \quad (90\% \text{ paired interval})$$

**Slower closed-route MSD growth is consistent with turns and returns.**

78 unclassified flights excluded. Nominal 90% cluster-bootstrap CIs; bands likely too narrow (Appendix).

# Does the open–closed $H$ gap change with altitude?



**Open:** higher  $H$  in lowlands

- Lowlands: no deviations for lift
- Mountains: deviations to known lift

**Closed:** higher  $H$  in mountains

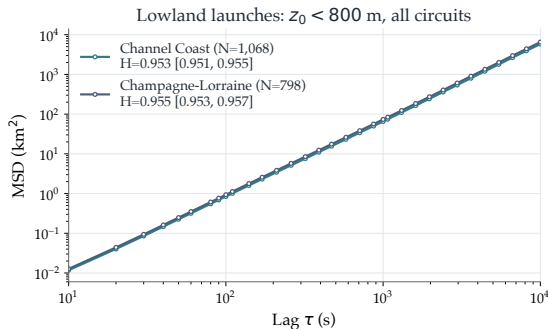
- Lowlands: turn back any time
- Mountains: way home follows ridges

Open > closed everywhere: gap 0.13 in lowlands, 0.02–0.05 in mountains.

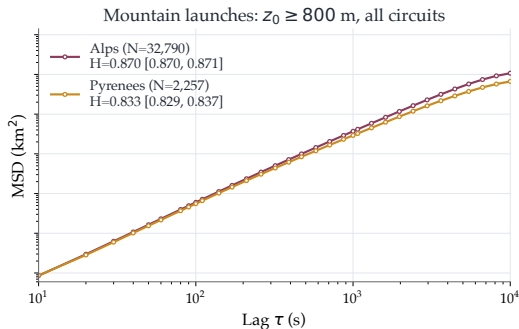
**Mountain terrain constrains both circuits, so the gap narrows.**

# How does $H$ differ by region?

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**Lowlands: almost equal**  
Channel Coast – Champagne-Lorraine  
 $-0.002$  [ $-0.005, 0.001$ ]



**Mountains: Alps above Pyrenees**  
Alps – Pyrenees  
 $0.037$  [ $0.033, 0.041$ ]

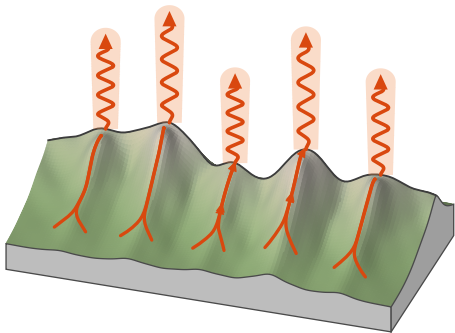
**$H$  separates the two mountain ranges more clearly than the two plains.**

# Why does $H$ separate the ranges but not the plains?

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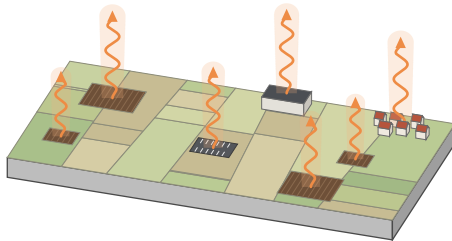
## Mountains: lift follows the slopes

Strong thermals climb sunny spurs to the crest.



## Lowlands: scattered triggers

Fields, asphalt, roofs: random spots, weaker lift.



**Mountain lift traces each range's relief; lowland lift is scattered at random.**

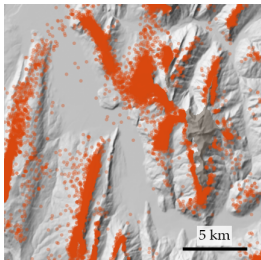
So the Alps and Pyrenees can differ in  $H$ , while two plains with similar trigger statistics cannot.

# Mountain lift follows ridges, lowland lift spreads aloft

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## Alps

17,603 flights at 800 m



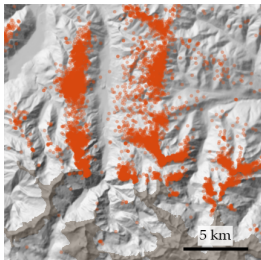
**H = 800 m**

49,105 climb crossings

[Open GIF](#)

## Pyrenees

2,634 flights at 800 m



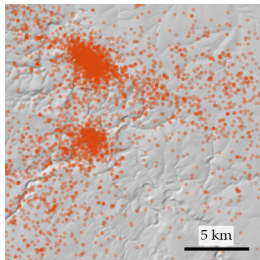
**H = 800 m**

10,557 climb crossings

[Open GIF](#)

## Channel Coast

1,939 flights at 800 m



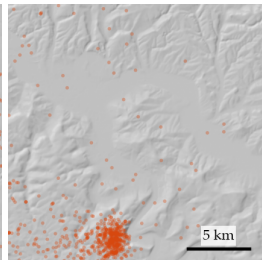
**H = 800 m**

4,140 climb crossings

[Open GIF](#)

## Champagne-Lorraine

474 flights at 800 m



**H = 800 m**

660 climb crossings

[Open GIF](#)

● climb crossing

■ terrain above the plane

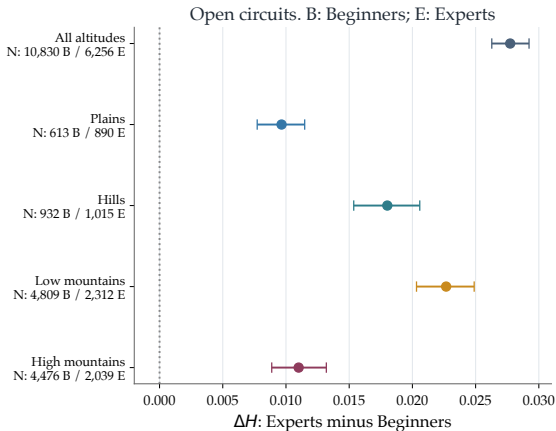
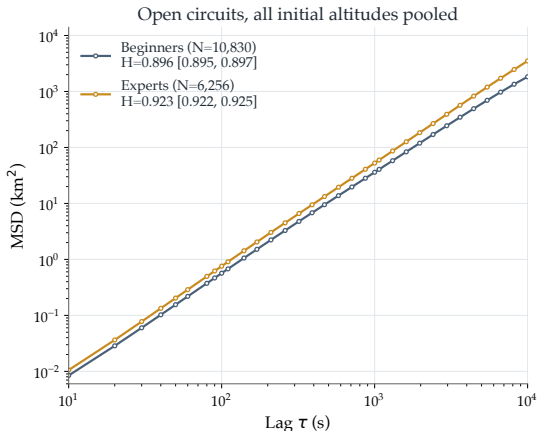
[Cell locations \(Appendix\)](#)

**Higher up, a random location may offer easier access to thermals in lowlands.**

This may explain similar Hurst exponents across lowlands and differences between mountain ranges.

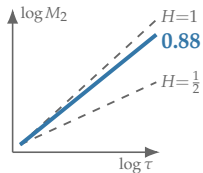
# Open circuits: expert–beginner $H$ gap by altitude

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Vilpellet et al. also found  $H_{\text{expert}} > H_{\text{beginner}}$   
with experts spending less time searching for thermals.

## Scaling



### Strongly superdiffusive

- Paragliders  $H = 0.88$
- Hang gliders  $H = 0.85$

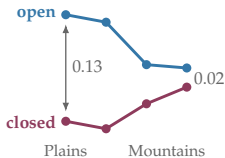
## Route and wing



Open routes spread faster

Experts spread faster

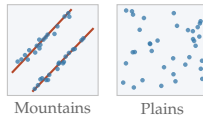
## Terrain



### Mountains narrow the gap

- Open-route  $H$  falls
- Closed-route  $H$  rises

## Lift geography



Mountains: ridge climbs

Plains: climbs spread out

Strongly superdiffusive everywhere; route and terrain set how strongly.

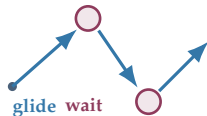
## Segmentation



Revise Vilpellet's HMM

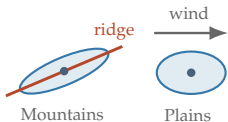


## Stochastic model



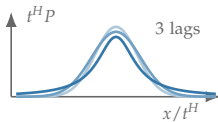
Which model fits best?

## Anisotropy



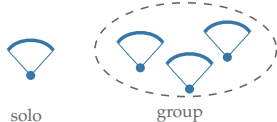
Ridge axis or wind

## Distribution scaling



Do distributions collapse?

## Solo vs group flight



Does group flight change  $H$ ?

Thank you  
for your attention!

# Appendix: clean the recorded flight

↔ slide 5 16

## 1 GNSS altitude

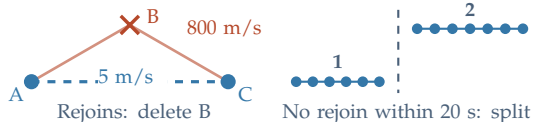
GNSS height → Trajectory  $z(t)$

Barometer → Cleaning checks

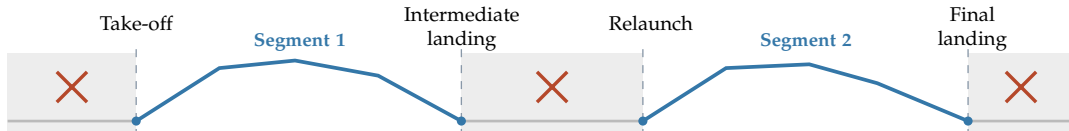
GNSS: absolute height without weather drift.  
Same high-frequency noise as the barometer.

## 2 Fix-level cleaning

One fix =  $(t, \varphi, \lambda, z)$ : time, latitude, longitude, height  
Delete an unreachable fix only if the track rejoins without it; otherwise split the flight.



## 3 Airborne interval



Trim before take-off and after landing.  
Stops are not labeled: we infer them from position and altitude.

## 4 Flight selection

Minimum flight thresholds

Time  $\geq 40$  min

1.13% excluded ( $N = 191,108$ )

Path  $\geq 20$  km

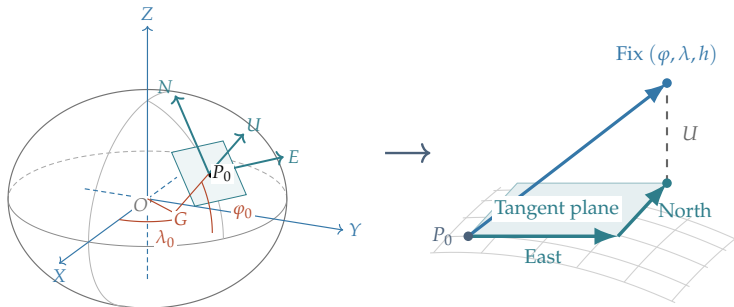
0.09% excluded ( $N = 188,942$ )

Altitude range  $\geq 600$  m

2.68% excluded ( $N = 188,768$ )

## 5 Coordinates

Latitude, longitude, height  $\rightarrow$  Earth-centred ( $X, Y, Z$ )  
 $\rightarrow$  local east-north distances in metres



Earth model: WGS84 ellipsoid (GPS reference)

$P_0$ : first trimmed fix; fixed for the flight.  $z = h_{\text{GNSS}}$ , not  $U$ .

## 6a Gaps: bridge or split

Gap  $g$  between consecutive fixes:

- $g \leq g_{\max}$ : bridge it.
- $g > g_{\max}$ : split into segments.

$$g_{\max} = \min(10\Delta t, \max(20\text{ s}, 2\Delta t))$$

Example:  $\Delta t = 5\text{ s}$ ,  $g_{\max} = 20\text{ s}$

Short gap  
15 s: bridge



Long gap  
30 s: split



● Raw fix  
(jittered)

● Grid point

○ Flagged as  
reconstructed

## 6b Time grid: jitter removal

Per segment: grid  $t_k = t_0 + k\Delta t$ , native  $\Delta t$ .

- Interpolate  $E, N, z$  at every  $t_k$ .
- No fix within  $\Delta t/2$  of  $t_k$ : flag it as reconstructed.



Keep a segment only if it lasts  $\geq 90\text{ s}$  and  $\leq 10\%$  of its grid is reconstructed.

**No segment kept: flight dropped.** 20,798 paragliders, 193 hang gliders: the largest loss.

## 7 Savitzky-Golay fit

Fit a cubic to  $E$ ,  $N$ ,  $z$  in a moving window, within each segment.

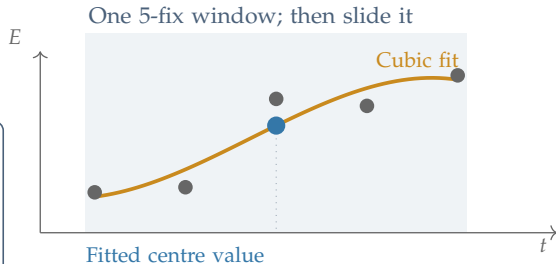
Window: odd,  $\geq 5$  fixes.

**Velocity and acceleration: derivatives of the fitted cubic**

$$v_i = P'_i(0), \quad a_i = P''_i(0), \quad u = t - t_i$$

Noise std, for an independent error  $\sigma$  per fix:

|                        | velocity                 | acceleration               |
|------------------------|--------------------------|----------------------------|
| Fitted cubic           | $0.95 \sigma / \Delta t$ | $0.53 \sigma / \Delta t^2$ |
| Fix-to-fix differences | $1.41 \sigma / \Delta t$ | $2.45 \sigma / \Delta t^2$ |



**Retained flights after all seven stages** (share of downloaded logs)

156,406 **paragliders** 84.1%

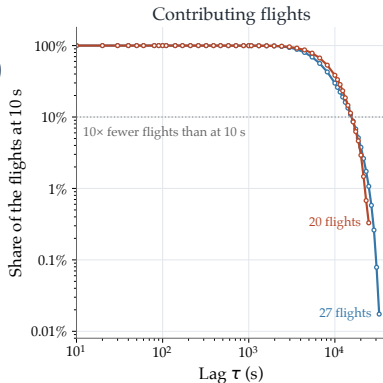
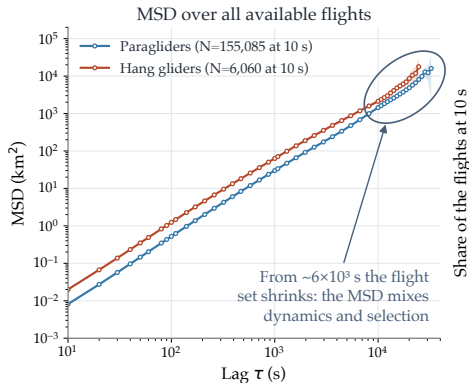
6,094 **hang gliders** 90.7%

Schematic; the cubic fit is exact for the synthetic fixes shown.

# Appendix: which flights contribute at each lag?

↔ slide 5 20

All available flights: a flight enters at lag  $\tau$  if one of its segments has  $\tau \leq 0.8 T_s$  ( $\geq 1/5$  of it left as origins for  $m_f$ ).



## Desiderata

- 1 The same flights at every lag
- 2 The longest possible lag range
- 3  $N \approx$  of the same order as at 10 s

At long lags only long flights remain, and their number falls steeply.

Any lag-dependent statistic then mixes dynamics with a changing set of flights.

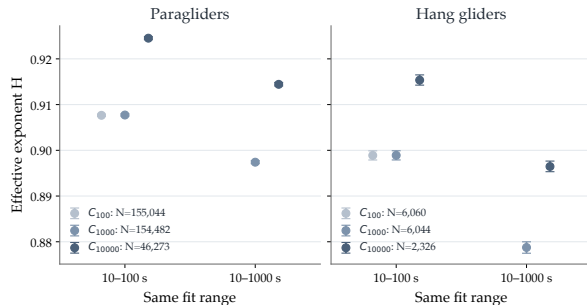
$C_{10000}$ : select once for the largest lag

$\tau_{\max} = 10,000 \text{ s} \leq 0.8 T_s$ :  $T_s \geq 12,500 \text{ s}$   
Only these segments enter, at every lag.

## Advantages of $C_{10000}$

- 1 Same flights and segments, 10 to 10,000 s, with no loss of flights beyond  $6 \times 10^3 \text{ s}$
- 2 Three decades of lag; at  $\tau_{\max}$  each segment still offers at least 251 origins for  $m_f$
- 3 Sample still large, same order as at 10 s:  
**46,273 paragliders** (30%)  
**2,326 hang gliders** (38%)

## Cost of selecting longer flights



Nested cohorts, identical fit ranges, paired bootstrap draws.

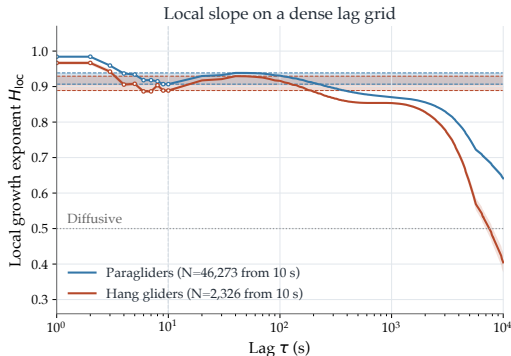
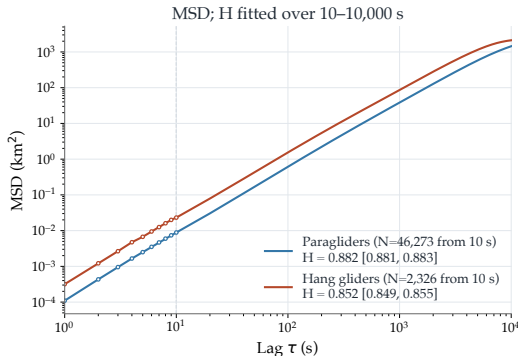
Long flights raise  $H$  by 0.016–0.018 on matched lag ranges; origins still decrease with lag.  
**About 2% in  $H$  buys the same flights at every lag over three decades.**

# Displacement growth from 1 s to 10,000 s

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$C_{10000}$ : changing support below 10 s; the same flights at every lag from 10 to 10,000 s.

$M_2(\tau) \propto \tau^{2H}$ :  $H = \frac{1}{2}$  diffusion  $H = 1$  ballistic motion



$$H_{\text{PG}} = 0.882 [0.881, 0.883] > H_{\text{HG}} = 0.852 [0.849, 0.855] \text{ over } 10\text{--}10,000 \text{ s}$$

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

## 1 Group flights

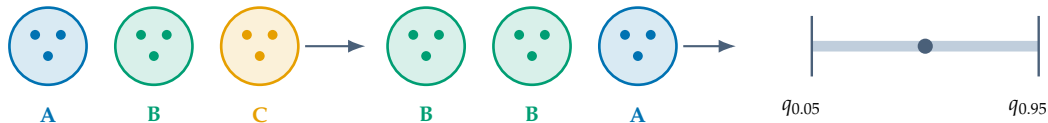
Take-off site  $\times$  calendar day.  
Aim: independent groups.

## 2 Resample groups

Sample with replacement.  
Keep each group together.

## 3 Recompute

1,000 replicates.  
Take the 5th–95th percentiles.



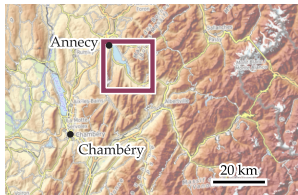
Current clusters are too small to absorb all shared flight conditions.

**Intervals likely underestimate uncertainty; clustering is still being refined**

# Appendix: locations of the 20 km cells

↔ slide 11 24

## Alps: Lake Annecy



Centre (WGS84)

$45.82374^{\circ}$  N

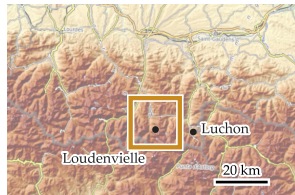
$6.22085^{\circ}$  E

Bounds (Lambert-93, m)

E 940000–960000

N 6520000–6540000

## Pyrenees: Louron Valley



Centre (WGS84)

$42.77884^{\circ}$  N

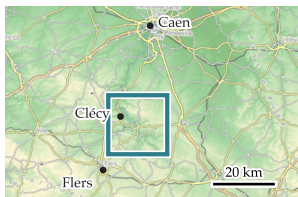
$0.43624^{\circ}$  E

Bounds (Lambert-93, m)

E 480000–500000

N 6180000–6200000

## Channel Coast: Suisse Normande



Centre (WGS84)

$48.88143^{\circ}$  N

$0.40990^{\circ}$  W

Bounds (Lambert-93, m)

E 440000–460000

N 6860000–6880000

## Champagne-Lorraine: Mouzon



Centre (WGS84)

$49.63141^{\circ}$  N

$5.07544^{\circ}$  E

Bounds (Lambert-93, m)

E 840000–860000

N 6940000–6960000

## Flight archive

FFVL Coupe Fédérale de Distance: IGC tracks, dates, declared sites, circuits and wing classes.

[parapente.ffvl.fr/cfd/liste](http://parapente.ffvl.fr/cfd/liste)   [delta.ffvl.fr/cfd/liste](http://delta.ffvl.fr/cfd/liste)

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## Maps and terrain

[OpenTopoMap](#): regional relief and contours (OSM / SRTM, CC-BY-SA).

[IGN / Géoplateforme](#): local Plan IGN backgrounds.

[IGN RGE ALTI](#): local terrain and derived crests.

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## Locator and analysis

[Natural Earth](#): France outline, public domain.

MSD, intervals and crossings: this thesis's prepared pipeline.

[Code, figure scripts and recorded source provenance](#)